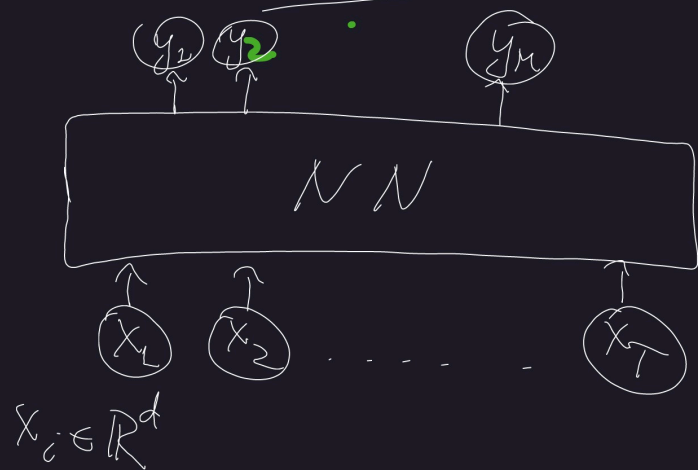
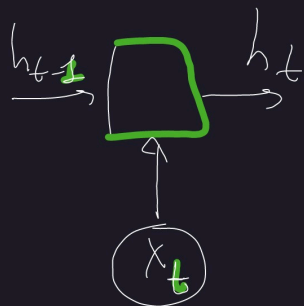


Recurrent Neural Networks and Attention



$$h_t = g(W^x x_t + W^h \cdot h_{t-1} + b)$$

$W^x \in \mathbb{R}^{\text{hidden_dim} \times \text{input_dim}}$ $b \in \mathbb{R}^{\text{hidden_dim}}$
 $W^h \in \mathbb{R}^{\text{hidden_dim} \times \text{hidden_dim}}$ $x_t \in \mathbb{R}^{\text{input_dim}}$



Given: $\nabla_{h_t} \mathcal{L}$, Find: $\nabla_{h_{t-1}} \mathcal{L}$

$$h_t = g(W^x x_t + W^h h_{t-1} + b) = z_t$$

$$d\mathcal{L} = \nabla_{h_t} \mathcal{L}^T dh_t = \underbrace{\nabla_{h_t} \mathcal{L}^T \frac{\partial g}{\partial z_t}}_{\nabla_{h_{t-1}} \mathcal{L}^T} \cdot W^h dh_{t-1}$$

$$\begin{aligned} \nabla_{h_{t-1}} \mathcal{L} &= (W^h)^T \frac{\partial g}{\partial z_t} \nabla_{h_t} \mathcal{L} = \\ &= \prod_{k=t}^T (W^h)^T \frac{\partial g}{\partial z_k} \nabla_{h_T} \mathcal{L} \end{aligned}$$

$$\begin{aligned} \|\nabla_{h_{t-1}} \mathcal{L}\| &\leq \\ &\leq \left(\prod_{k=t}^T \|W^h\| \cdot \left\| \frac{\partial g}{\partial z_k} \right\| \right) \|\nabla_{h_T} \mathcal{L}\| \end{aligned}$$

Exploding grating: gradient clipping

$$\frac{\partial \mathcal{L}}{\partial \theta} = \begin{cases} g, & \text{if } \|g\|_2 \leq N = 10^{10} \\ \frac{N}{\|g\|_2} g, & \text{if } \|g\|_2 > N \end{cases}$$

Vanishing grating:

- use ReLU, leaky ReLU, ...
- use orthogonal W^h
 - init. W^h with orth
 - add regularizer $+\lambda \|W^T W - I\|_F^2$
 - use orth. reparameterization
- use LSTM, GRU

$$f(x) \rightarrow \min_{x > 0}$$

$$a \leq x \leq b$$

$$x = a + (b-a) \underbrace{\sigma(y)}_{\substack{\uparrow \\ (0, 1)}}$$

$$X \geq 0$$

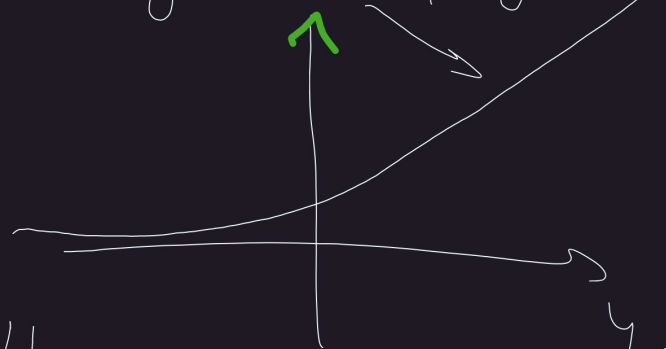
$$X = Y^T Y$$

$$X \in \mathbb{R}^{h \times h} \quad Y \in \mathbb{R}^{h \times u}$$

$$x = \exp(y), \quad y \in \mathbb{R}$$

$$f(\exp(y)) \rightarrow \min_y$$

$$x = \log(1 + \exp(y))$$



Orthogonal: $X = \expm(Y - Y^T)$

LSTM (Long Short Term Memory)

C_t - long term memory

h_t - short term memory

$$g_t = g(W_x^g x_t + W_h^g h_{t-1} + b^g)$$

$$i_t = \sigma(W_x^i x_t + W_h^i h_{t-1} + b^i) \in (0, 1)$$

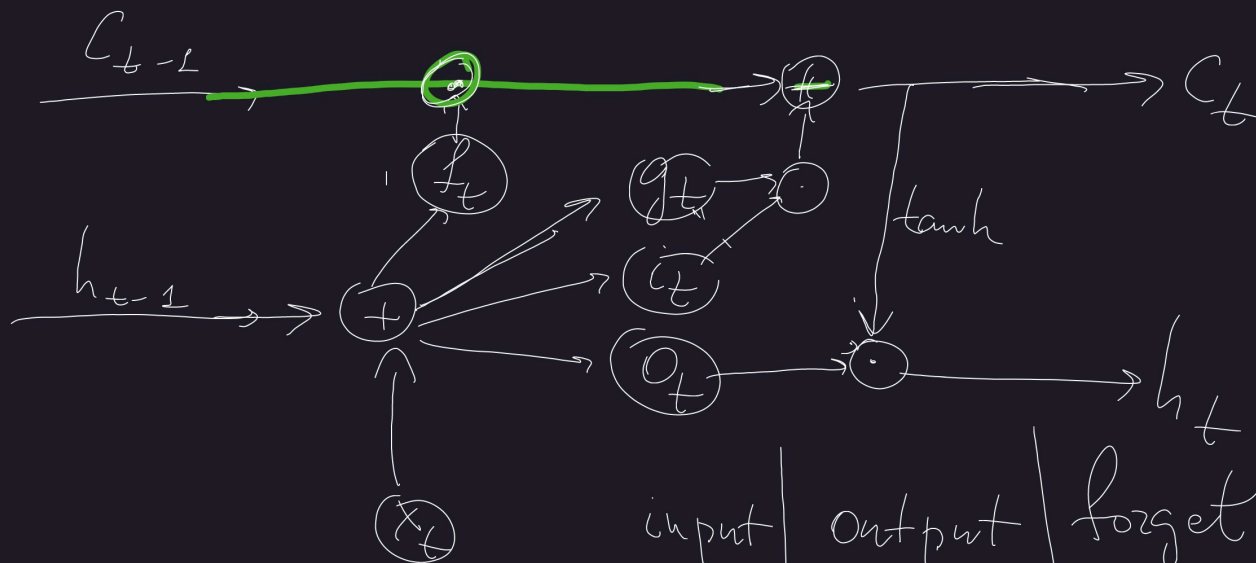
$$o_t = \sigma(W_x^o x_t + W_h^o h_{t-1} + b^o)$$

$$f_t = \sigma(W_x^f x_t + W_h^f h_{t-1} + b^f)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot g_t$$

$$h_t = o_t \odot \tanh(C_t)$$

Init: $b^f \gg 1$



$$f_t \approx 0$$

$$\frac{\partial C_t}{\partial C_{t-1}} \approx I$$

Behaviour of f LSTM		
input	output	forget
+	+	-
+	-	+
-	+	+
-	-	-

standard RNN
 store information into memory
 load inf. from memory
 erase memory

GRU (Gated Recur. Unit)

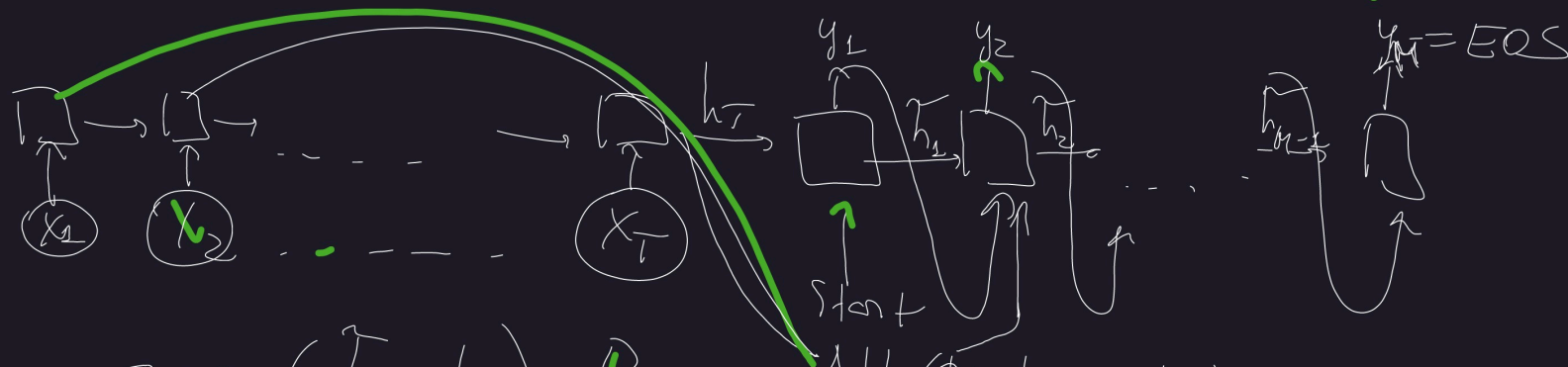
$$z_t = \sigma(W_x^z x_t + W_h^z h_{t-1} + b^z) - \text{reset gate}$$

$$u_t = \sigma(W_x^u x_t + W_h^u h_{t-1} + b^u) - \text{update gate}$$

$$g_t = g(W_x^g x_t + W_h^g (h_{t-1} \odot z_t) + b^g)$$

$$h_t = (1 - u_t) \odot h_{t-1} + u_t \odot g_t \quad \text{Init. } b^u \gg 1$$

$aebddcaec \rightarrow aebddcaec$ $aebddcaec$



$$\text{score}(\vec{h}, h) \in \mathbb{R}$$

$$\text{Att.}(\vec{h}_1, h_1, \dots, h_T)$$

$$d_{it} = \frac{\exp(\text{score}(h_i, \vec{h}_{t-1}))}{\sum_{j=1}^T \exp(\text{score}(h_j, \vec{h}_{t-1}))} \quad \forall i=1, \dots, T$$

$$\text{Att.}_t = \sum_{j=1}^T d_{jt} h_j$$

$$\text{score}(h, \tilde{h}) : \begin{array}{l} \textcircled{1} \quad h^T \tilde{h} \\ \textcircled{2} \quad \frac{h^T \tilde{h}}{\sqrt{\dim(h)}} \end{array}$$

$$\textcircled{3} \quad w^T \tanh(W_1 h + W_2 \tilde{h} + b)$$

$$h_i, \tilde{h}_j \sim \mathcal{N}(0, 1)$$

$$h^T \tilde{h} = \sum_i h_i \tilde{h}_i$$

$$E(h^T \tilde{h}) = \sum_i \underset{0}{E h_i} \cdot \underset{0}{E \tilde{h}_i} = 0$$

$$\begin{aligned} \text{Var}(h^T \tilde{h}) &= \sum_i \underset{1}{\text{Var}(h_i)} \underset{1}{\text{Var}(\tilde{h}_i)} \\ &= \dim(h) \end{aligned}$$

